

Predictive KV Cache Amortization for Edge Language Model Inference on Mobile Devices

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Abstract—Context-aware language model inference on mobile devices requires processing environmental prompts before generating responses, with prefill cost scaling linearly in prompt length. We present Context-Aware Predictive KV-Cache Priming (CAP-KVC), which amortizes this cost by using environmental sensor signals—GPS, Bluetooth Low Energy proximity, and time of day—to predict the user’s likely context and pre-compute key-value cache tensors during idle periods. The pre-computed cache is restored in 2.74 ms on average, bypassing the context prefill entirely. On a OnePlus 11R with Qwen2.5-0.5B-Instruct (Q4_K_M), CAP-KVC achieves a $4.24\times$ end-to-end speedup at approximately 500 tokens. Native-side `std::chrono` instrumentation reveals a $40.78\times$ prefill-specific speedup: 9 intent tokens (343 ms) versus 439 tokens for RAG_INLINE (14001 ms) at the 500-token tier. The system is a complete Android application with a C++/JNI backend based on `llama.cpp`, a bounded KV cache manager supporting three simultaneous states within 118.7 MB native heap, and sensor-driven context routing with hysteresis-gated drift detection. A proof-of-concept multimodal intent layer combining surface electromyography and gaze tracking through late fusion is included; its evaluation uses synthetic data and does not constitute clinical validation. All measurements, statistical tests, and limitations—including single-device evaluation—are reported.

Index Terms—Key-value cache, edge inference, predictive caching, large language model, mobile systems, sensor fusion, Android, augmentative and alternative communication.

I. INTRODUCTION

For individuals with severe speech and motor impairments resulting from conditions such as amyotrophic lateral sclerosis (ALS) or high-level spinal cord injury, augmentative and alternative communication (AAC) devices serve as the primary lifeline for social interaction and autonomy [1]. Conversational turn-taking fundamentally requires sub-second system responsiveness; latency exceeding 1–2 seconds severely degrades dialogue flow, exhausts motor effort, and leads to communication breakdown.

Recent advances in 4-bit weight quantization [4], [5] and mobile execution engines [2] have enabled sub-billion-parameter language models to execute entirely on-device, offering personalized, privacy-preserving phrase completion and intent expansion. To generate contextually relevant utterances, modern AAC pipelines prepend environmental context prompts—capturing the user’s location, daily schedule, companion identity, or clinical state—to the user’s sparse input tokens before inference.

However, contextual framing introduces a severe computational penalty. In autoregressive transformer architectures, processing prompt tokens (the *prefill* phase) scales as $O(N)$ with context sequence length [30] and dominates end-to-end latency on resource-constrained mobile processors. On an arm64 mobile SoC, evaluating a modest 500-token contextual prompt can require 14 seconds of continuous matrix multiplication before the first token is emitted, rendering turn-taking impossible.

In server-side environments, this prefill latency is typically amortized through reactive KV cache reuse, such as prefix caching in vLLM [11], radix tree indexing in SGLang [10], and retrieval-aware KV caching in RAGCache [12]. However, these approaches are fundamentally *reactive*: they require an inference request to arrive, match the request against an in-memory index of past queries, and incur the full $O(N)$ prefill penalty on the first request to any new context. On single-user mobile AAC devices, where interactions are episodic and separated by substantial idle intervals, reactive caching fails to mitigate the first-turn latency spike.

To address this challenge, we introduce **Context-Aware Predictive Key-Value Cache Priming (CAP-KVC)**. Rather than waiting for a user query to trigger context processing, CAP-KVC exploits ambient environmental sensor signals—GPS coordinates, Bluetooth Low Energy (BLE) beacon proximity, and time of day—to proactively predict the user’s upcoming semantic context during device idle periods. The predicted context is pre-evaluated through the language model backend, and the resulting key-value attention tensors are serialized to local storage. When the user subsequently initiates an intent, the matching KV cache is restored in under 3 ms, completely bypassing the context prefill phase.

Contributions:

- 1) **Predictive KV Cache Priming Architecture:** We formulate a proactive caching paradigm for edge devices that decouples prompt prefill from user interaction by predicting environmental context transitions. On a OnePlus 11R smartphone with Qwen2.5-0.5B-Instruct (Q4_K_M), native-side prefill decomposition confirms a $40.78\times$ prefill-specific speedup at $N \approx 500$ tokens (343 ms vs. 14001 ms) and a $4.24\times$ end-to-end speedup (4574 ms vs. 19408 ms). In matched-context evaluation (~ 50 tokens), prefill speedup is $5.89\times$ (343 ms vs. 2021 ms, Welch’s $t(29.0) = -2.71$, $p = 0.011$, Cohen’s

$d = 0.70$), while cache restoration overhead is negligible (2.74 ms).

- 2) **Edge-Native Android Implementation:** We present a production-grade Android implementation featuring a scalar-only JNI boundary with 13 native methods (8 core inference/cache operations, 1 lifecycle cleanup, and 4 high-resolution native telemetry getters), reentrant-lock concurrency protecting global native state, and an LRU cache manager maintaining a bounded residency of $k = 3$ active KV states within an 118.7 MB native heap budget.
- 3) **Sensor Routing & Threat Model Hardening:** A closed-form convex scoring function combines temporal, GPS, and BLE signals with reliability-adaptive weighting. To address physical-world threat models, we integrate a dedicated `LocationValidator` providing mock-location detection, fix-staleness bounding (≤ 120 s), and accuracy sanity gating ($2m \leq \text{acc} \leq 500m$), alongside hysteresis-gated drift detection ($\Delta = 0.10$) to prevent cache thrashing.
- 4) **Multimodal Intent & Controlled Trace Evaluation:** A proof-of-concept multimodal intent layer combines surface electromyography (sEMG) with gaze tracking via late fusion (macro F1 = 0.9963 on synthetic data). We rigorously evaluate context prediction across 100 simulated daily traces (6,400 total interactions) with realistic independent noise models, demonstrating 98.8% Hit@1, 100% Hit@3, and zero cold misses under full sensor fusion, while characterizing degradation under high noise (89.3% Hit@1) and adversarial GPS spoofing (67.7% Hit@1 with zero fallback context switches).

A. Research Questions

- **RQ1** (Prefill Amortization & Latency Decomposition): To what extent does predictive KV cache priming reduce prefill and end-to-end inference latency relative to inline processing on mobile SoCs, both at matched context sizes and across context length scaling?
- **RQ2** (Prediction Accuracy & Robustness): What prediction accuracy (Hit@1, Hit@3, cold miss rate) does multi-modal sensor fusion achieve across longitudinal daily traces, and how robust is the routing engine to sensor dropout, high environmental noise, and adversarial spoofing?
- **RQ3** (System Overhead & Concurrency): What native heap overhead and cache restoration costs are required to sustain bounded multi-state KV cache residency, and can drift detection prevent oscillatory cache thrashing?
- **RQ4** (Multimodal Intent Fusion): Can a lightweight late fusion architecture successfully combine heterogeneous biosignal embeddings (sEMG and gaze) for intent classification under controlled conditions?

II. RELATED WORK

A. KV Cache Optimization & Prefix Reuse

Token-level cache compression. Several techniques prune KV cache tensors within active sequences. Scissorhands [7]

and H₂O [8] exploit attention persistence to evict low-importance tokens. SnapKV [21] identifies consistent attention heads for cluster-level pruning. StreamingLLM [9] retains initial “attention sink” tokens alongside a rolling local window for infinite sequence lengths. KVQuant [25] quantizes KV tensors to low bit-widths. These methods operate intra-sequence and are orthogonal to state-level prefill caching.

Prompt caching and prefix sharing. Server-side inference systems increasingly exploit prefix redundancy across queries. SGLang’s RadixAttention [10] organizes previously evaluated prompt tokens into a radix tree for multi-turn conversational reuse. vLLM’s PagedAttention [11] segments KV caches into virtual blocks to facilitate shared-prefix memory management. Prompt Cache [22] precomputes and saves attention states for modular prompt segments, reducing time-to-first-token (TTFT) on cloud GPUs. CacheBlend [23] extends prefix caching to non-prefix-aligned text chunks. RAGCache [12] organizes retrieval knowledge into a multi-tier cache hierarchy across host memory and GPU VRAM.

Predictive priming vs. reactive caching. A critical distinction separates CAP-KVC from server-side prefix caching: systems like SGLang, vLLM, and RAGCache are fundamentally *reactive*. They assume an incoming request stream and match prompt prefixes against previously evaluated queries. Under reactive caching, the *first* interaction in any new context necessarily incurs the full $O(N)$ inline prefill latency. In contrast, CAP-KVC is an *ambient, predictive* edge architecture. By monitoring physical-world environmental sensors (GPS, BLE, time) during user idle periods, CAP-KVC anticipates context shifts and primes the cache *before* any communicative intent is triggered, ensuring zero prefill penalty on the critical first conversational turn.

B. Edge Deployment of Language Models

Post-training quantization to 4-bit [4], [5] and architecture optimizations [27] enable sub-billion-parameter models on smartphone SoCs. Frameworks such as `llama.cpp` [2] and MLC-LLM [6] provide ARM NEON-optimized kernels for mobile execution. PowerInfer-2 [24] explores heterogeneous NPU/CPU scheduling for large models, while Alizadeh et al. [28] and FlexGen [29] investigate flash memory offloading. However, when prompts contain substantial contextual framing, the $O(N)$ prefill phase remains the dominant contributor to user-perceived delay.

C. AAC and Multimodal Biosignal Interfaces

AAC devices serve individuals with severe motor and vocal impairments [1]. Integrating language models for word completion and contextual expansion [17] provides significant communicative bandwidth improvements, provided response latency remains within conversational tolerance. Surface EMG gesture classification [15] and gaze tracking via MediaPipe [16] provide non-invasive input modalities; late fusion architectures are widely favored for combining heterogeneous sensor streams with disparate temporal and spatial resolutions [18].

D. Positioning

CAP-KVC unifies predictive physical-world sensing with edge-native KV cache amortization. While prior KV caching frameworks focus on multi-tenant server workloads with text-based matching, CAP-KVC demonstrates that ambient sensor signals on mobile devices provide sufficient predictive power to prime on-device language models proactively, achieving sub-second edge responsiveness for assistive communication.

III. METHODOLOGY

CAP-KVC amortizes the cost of contextual prompt ingestion on mobile hardware by executing it as a background process prior to user interaction. Fig. 1 presents the system architecture.

A. Problem Formulation

Consider a user u operating an AAC device in a physical environment characterized by temporal, spatial, and proximity signals. Let $C = \{c_1, c_2, \dots, c_M\}$ denote a finite set of pre-configured semantic contexts, each associated with a textual system prompt p_i of length N_i tokens. In conventional retrieval-augmented generation (RAG) pipelines, the system prompt p_i is concatenated with the user's intent tokens and submitted to the language model at inference time, incurring an $O(N_i)$ prefill cost that directly adds to user-perceived latency.

CAP-KVC reformulates this as a scheduling problem: given a prediction of the user's most likely context $\hat{c}$ based on environmental signals, execute the prefill of $p_{\hat{c}}$ during idle periods and persist the resulting key-value (KV) cache tensors to persistent storage. At inference time, the system restores the pre-computed KV cache and appends only the user's intent tokens, reducing the inference-time cost from a prompt-length-dependent prefill to a cache restoration operation whose cost is independent of the original prompt length N_i .

B. System Execution Lifecycle

The system operates through a five-stage pipeline, where the first three stages execute asynchronously during background idle periods and the final two stages execute at user-interaction time:

- 1) **Sensor Acquisition.** The system concurrently polls available hardware sensors—GPS, Bluetooth Low Energy (BLE), and the system clock—to construct an immutable `SensorSnapshot` representing the user's current environmental state.
- 2) **Deterministic Context Routing.** The snapshot is evaluated by a `StateRouter` against all registered semantic contexts using a convex scoring function (Section III-D). The top- k highest-scoring contexts are selected for cache residency, where k is bounded by a device-specific RAM constraint (Section III-E).
- 3) **Background Cache Priming.** For each selected context not already resident in memory, the system executes a background prefill of its system prompt through the language model, serializes the resulting KV cache tensors to persistent storage, and registers the context as resident.

- 4) **Resident Inference.** Upon user interaction, the pre-computed KV cache for the predicted context is restored from storage into native memory. The user's intent tokens are then appended and decoded, bypassing the context prefill entirely.
- 5) **Drift Correction.** An asynchronous background daemon periodically re-evaluates environmental scores to detect changes in the user's physical context and triggers cache replacement when a new context scores sufficiently higher than the weakest resident context (Section III-F).

C. Multimodal Intent Classification

User intent is derived from two complementary input modalities, fused to produce a single classification decision that is then formatted as a textual prompt for the language model.

1) *Surface Electromyography (sEMG):* A facial sEMG wearable provides 16-channel signals sampled continuously. Raw signals are segmented into fixed-length windows and processed by a CNN-LSTM classifier that extracts a 64-dimensional embedding vector. The CNN block applies three sequential Conv1d–BatchNorm–ReLU–MaxPool layers for temporal feature extraction, followed by a two-layer LSTM for temporal context modeling. The final hidden state is projected to a 64-dimensional space and L2-normalized to produce the EMG embedding.

2) *Gaze Tracking:* A front-facing camera feeds a MediaPipe Face Mesh model [16] that extracts 468 facial landmarks at frame rate. From these, the system computes a 5-dimensional gaze feature vector from the displacement between the nose tip (landmark 4) and the eye center (midpoint of landmarks 33 and 362):

$$\mathbf{g} = [d_x, d_y, |d_x|, |d_y|, \sqrt{d_x^2 + d_y^2}] \quad (1)$$

where d_x and d_y are the horizontal and vertical displacements in normalized image coordinates. This vector captures both directional and magnitude information while remaining invariant to face scale. A dwell-based finite state machine maps sustained gaze fixation (exceeding 400ms on a target region) to one of five discrete AAC intent classes: *confirm*, *reject*, *scroll*, *select*, and *call-help*.

3) *Late Fusion Architecture:* The EMG embedding $\mathbf{e} \in \mathbb{R}^{64}$ and the gaze vector $\mathbf{g} \in \mathbb{R}^5$ are fused through a late fusion architecture. The gaze vector is first projected to a 64-dimensional space via a two-layer MLP, then concatenated with the EMG embedding to produce a 128-dimensional joint representation. A final two-layer classifier maps this to 5-class intent logits:

$$\hat{y} = \text{MLP}_{64 \rightarrow 5}(\text{ReLU}(\text{MLP}_{128 \rightarrow 64}([\mathbf{e}; \text{proj}(\mathbf{g})]))) \quad (2)$$

Late fusion was selected over a cross-attention alternative based on a pre-specified decision criterion: cross-attention must exceed late fusion F1 by more than 3% (absolute) to justify its additional computational and parameter overhead. The ablation results (Section VI-C) did not meet this threshold.

CAP-KVC System Architecture

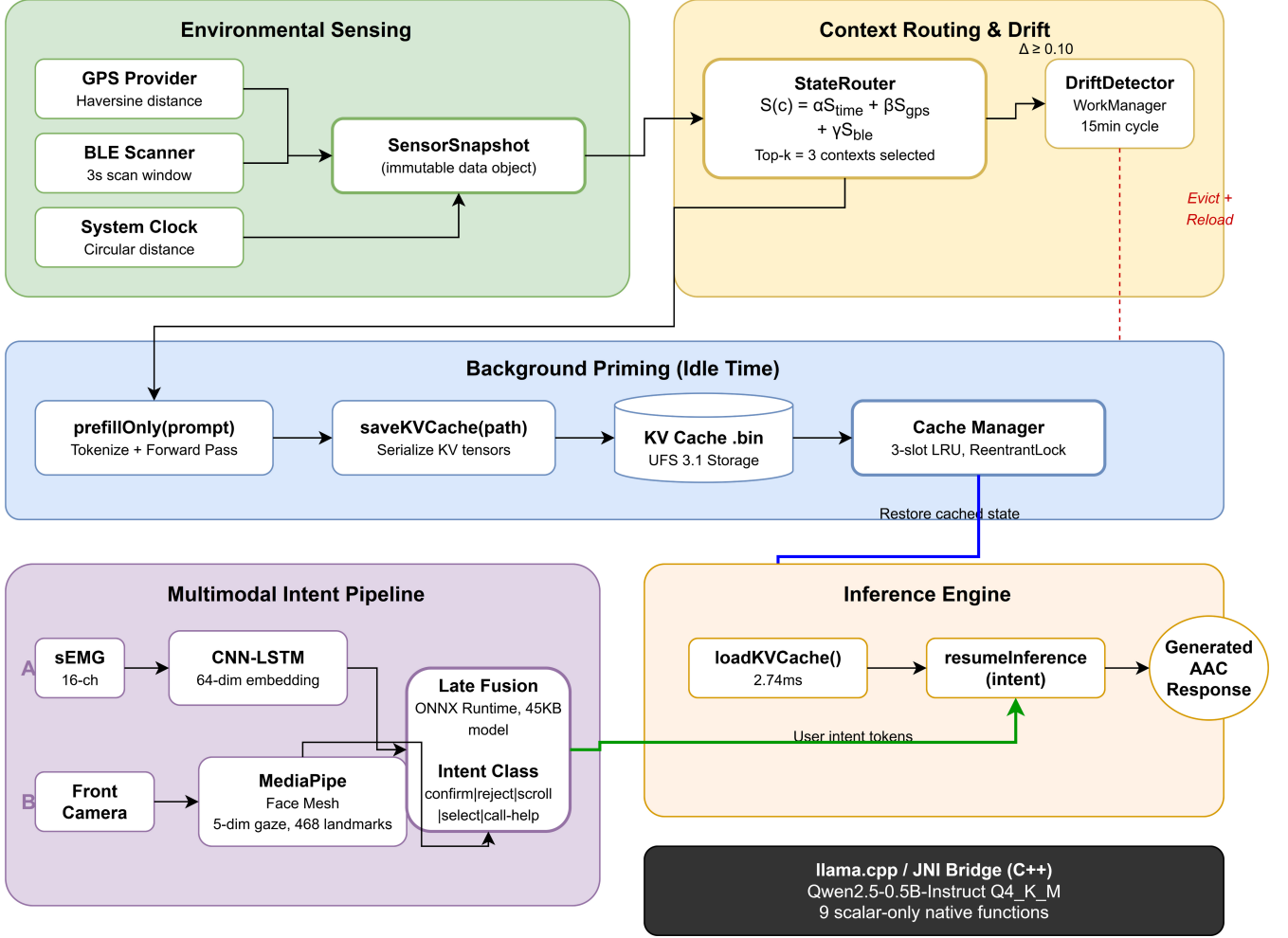


Fig. 1. CAP-KVC system architecture. Environmental sensors (blue) feed a deterministic scoring function that selects contexts for background priming (green). The multimodal intent pipeline (purple) combines sEMG and gaze inputs via late fusion. At inference time (orange), the pre-computed KV cache is restored and only intent tokens are processed by the llama.cpp/JNI backend. A periodic drift detector re-evaluates context relevance with hysteresis gating ($\Delta \geq 0.10$).

D. Deterministic Context Scoring

We employ a deterministic, closed-form scoring function rather than a learned predictor to ensure bounded CPU and battery consumption during background polling. A semantic context c_i is scored against the current sensor snapshot as a convex combination of three environmental similarity scores:

$$S(c_i) = \alpha \cdot S_{\text{time}} + \beta \cdot S_{\text{gps}} + \gamma \cdot S_{\text{ble}} \quad (3)$$

where $\alpha + \beta + \gamma = 1$. The weights are derived from dynamic reliability estimates of each sensor signal (Section III-D4).

1) *Temporal Scoring*: The temporal similarity S_{time} applies a Gaussian kernel to the circular distance between the current time and the context's expected activation time:

$$S_{\text{time}} = \exp\left(-\frac{d_{\text{circ}}(t, t_i)^2}{2\sigma^2}\right) \quad (4)$$

where d_{circ} handles the 24-hour wraparound and $\sigma = 2.0$ hours accommodates typical schedule variability.

2) *Spatial Scoring*: The spatial similarity S_{gps} applies an exponential decay to the Haversine great-circle distance between the current GPS coordinates and the context's anchor location:

$$S_{\text{gps}} = \exp(-\lambda \cdot d_{\text{haversine}}) \quad (5)$$

with $\lambda = 0.1$ km (decay constant), establishing a contextual boundary of approximately 100 meters. At $d = 100$ m, the score decays to ≈ 0.37 ; at $d = 1$ km, it falls to near zero.

3) *BLE Proximity Scoring*: The BLE similarity S_{ble} evaluates the overlap between currently detected Bluetooth devices and a context's registered device topology, with RSSI values mapped through a sigmoid function.

4) *Dynamic Reliability Normalization*: The weights α , β , and γ are computed dynamically from raw sensor reliability estimates rather than being statically configured. Time maintains a fixed baseline weight $w_t = 0.4$, guaranteeing that the normalization denominator $w_t + w_g + w_b$ is always

positive even when both GPS and BLE are unavailable. GPS reliability w_g decays exponentially with reported accuracy ($w_g = \exp(-a/50.0)$ where a is the accuracy in meters), reflecting the degradation of location precision in indoor environments. BLE reliability w_b is computed from the mean sigmoid-transformed RSSI of detected devices, centered at -70 dBm. If GPS is unavailable ($w_g = 0$) or no BLE devices are detected ($w_b = 0$), the corresponding weight is zero and the remaining weights are renormalized to preserve the convex property $\alpha + \beta + \gamma = 1$. The value $w_t = 0.4$ was selected as an engineering heuristic to ensure meaningful temporal scoring under all sensor configurations; no systematic sensitivity analysis was performed. Contexts scoring below a dead-state threshold of 0.01 are excluded from routing consideration.

E. KV Cache Residency Management

The system configures a finite set of $M = 5$ pre-configured semantic contexts representing typical daily AAC living environments: `home_morning`, `home_evening`, `hospital_ward`, `therapy_room`, and `caregiver_visit`. The cache manager maintains a resident subset of at most k active KV states simultaneously in native memory ($k = 3$ in our deployment).

Residency budget rationale ($k = 3$). The selection of $k = 3$ balances prompt coverage against edge device memory pressure. At 4-bit quantization (Q4_K_M) for Qwen2.5-0.5B with a maximum sequence allocation of 512 tokens per slot, each resident state consumes approximately 39.6 MB of native memory, yielding a total footprint of 118.7 MB for $k = 3$. Modern Android runtime configurations typically impose per-process virtual memory thresholds between 192 MB and 256 MB before triggering background trim callbacks. A residency budget of $k = 3$ leaves ample headroom for the user interface, camera frame buffers, and transient inference scratchpads, completely avoiding OS-level `onTrimMemory()` or low-memory killer (LMK) evictions. Functionally, maintaining 3 out of 5 states covers 60% of all configured contexts simultaneously resident, ensuring that transitions between frequent collocated states (e.g., morning and evening home routines, which share spatial coordinates but differ temporally) do not require persistent storage deserialization.

When a new context requires loading and all k slots are occupied, the least-recently-used (LRU) state is evicted. Eviction is protected by an atomic reference-counting mechanism: a state undergoing active inference holds an in-flight reference that blocks eviction until text generation completes. The eviction worker polls with a 10 ms backoff until the reference count drains to zero before reassigning the native slot.

F. Asynchronous Drift Detection

Environmental sensors predict physical location but cannot detect whether the conversation topic has diverged from the cached context. A periodic background daemon (executing every 15 minutes via the Android WorkManager API) re-evaluates context scores using a lightweight sensor snapshot and compares them against currently resident states.

To prevent cache thrashing under noisy sensor conditions, state replacements are gated by a hysteresis margin: a candidate context must exceed the weakest resident state’s score by at least $\Delta = 0.10$ before a cache swap is authorized. This prevents oscillatory eviction-reload cycles that would impose unnecessary I/O and prefill costs.

A separate offline characterization study evaluates the sensitivity of *semantic* topic-shift detection—using SentenceBERT centroid distances on annotated DailyDialog conversations [19]—to the sliding window size k used for centroid smoothing (Section VI-F). This offline study is distinct from the deployed `DriftDetector`, which operates on sensor-score deltas rather than semantic embeddings. The offline ablation characterizes detection sensitivity as a function of k on available annotated data; the deployed system’s behavior is governed by the hysteresis margin Δ described above.

G. Inference Fallback

If a user triggers an intent before the background daemon has successfully pre-loaded any relevant context (a “cold miss”), the system employs a two-tier fallback pipeline. Tier 1 executes an immediate in-memory lookup of pre-configured generic responses and presents the result via text-to-speech output, maintaining immediate conversational responsiveness. Tier 2 initiates an asynchronous standard RAG prefill on the inference backend, allowing subsequent user intents to be upgraded to contextual inference once the prefill completes.

IV. IMPLEMENTATION

A. Platform and Runtime

AACBridge targets `arm64-v8a` devices running Android 12+. The inference backend builds on `llama.cpp` [2], using Qwen2.5-0.5B-Instruct [3] quantized to Q4_K_M via GGUF. Pre-compiled native libraries ship inside the APK. The Kotlin layer communicates with C++ through JNI; all dependency injection is manual (no Hilt/Dagger/Koin) to minimize startup latency.

B. JNI Boundary Contract

The JNI layer exposes 13 native functions in a single C++ source file (`llama_jni.cpp`, 801 lines), divided into two tiers: (1) nine core operational functions comprising eight application-facing primitives plus one lifecycle cleanup (`release`); and (2) four high-resolution telemetry getters (`getLastPrefillMs`, `getLastGenMs`, `getLastPromptTokens`, `getLastGenTokens`) designed for native phase decomposition. A strict constraint governs this boundary: **all arguments and return values are JNI scalar types** (`jstring`, `jint`, `jdouble`, `jboolean`). No Java objects, arrays, or callbacks cross the boundary, eliminating JNI global references and GC root pinning.

The operational functions fall into three groups: *lifecycle* (`initializeBackend`, `initializeModel`, `release`); *KV cache operations* (`saveKVCache`, `loadKVCache`, `clearKVCache`); and *inference* (`runInference`, `prefillOnly`, `resumeInference`).

The distinction between `runInference` and `resumeInference` is critical: the former clears session history and processes from position zero, while the latter preserves the loaded token history and appends at position n_{past} , preventing position-ID collisions that would corrupt the attention computation.

C. Concurrency Model

The C++ backend maintains shared global state: a `llama_context` pointer and a `session_tokens` vector for KV cache consistency. A single `ReentrantLock` (the “engine lock”) guards all JNI call sequences that depend on consistent session state. The lock is held across both `loadKVCache` and `resumeInference` within a single acquisition, since releasing between calls would let another thread mutate `session_tokens`. Separate per-state Kotlin coroutine `Mutex` instances protect cache residency metadata without blocking unrelated operations.

D. KV Cache Priming Lifecycle

Context priming executes three phases per semantic state: (1) *prefill*—the engine lock is acquired and `prefillOnly(prompt)` populates the native KV cache without entering the generation loop; (2) *serialize*—while still holding the lock, `saveKVCache(tmpPath, 0)` writes KV tensors to a temporary file (slot 0 is mandatory since `prefillOnly` decodes into slot 0 via `llama_batch_get_one`); (3) *atomic commit*—after releasing the lock, the temporary file is atomically renamed to the final `.bin` path, leveraging `ext4`’s `atomic rename()`. A model-readiness gate (`AtomicBoolean`) prevents priming before native context initialization.

E. KV Cache Residency and Eviction

The cache manager maintains $k = 3$ native sequence slots as a concurrent queue. When all slots are occupied, the least-recently-accessed state is evicted: the victim is marked inactive, the manager polls until its reference count drains to zero (10ms delay to avoid busy-spinning), and the slot is returned to the pool. Reusing a slot via `loadKVCache` implicitly overwrites prior KV contents after clearing via `llama_memory_seq_rm`.

F. Sensor Integration & LocationValidator

Two cooperating components handle background sensing. **ActiveSweep** is a coroutine-based collector that concurrently acquires GPS fixes and BLE topology (3000ms `BluetoothLeScanner` window), combining both into a validated `SensorSnapshot`. **DriftDetector** is a `CoroutineWorker` polling every 15 minutes via `WorkManager`; it uses a BLE-blind snapshot to conserve battery, relying on GPS and time signals—methodologically valid because both candidate and resident states are scored against the same BLE-blind data.

To guard against corrupted or malicious telemetry, all GPS fixes pass through a dedicated `LocationValidator`

before entering the routing pipeline. The validator enforces three rejection criteria: (1) *Mock-location rejection*: checks `Location.isMock()` (API 31+) or `Location.isFromMockProvider()` to reject simulated coordinates; (2) *Fix staleness*: discards fixes older than 120 seconds ($2 \times$ the sweep interval) using elapsed real-time clocks; and (3) *Accuracy plausibility*: rejects fixes reporting suspicious super-precision (< 2.0 m, typical of mock injectors) or extreme imprecision (> 500 m).

G. Threat Model and Attack Surface Boundary

Edge-deployed AAC systems operating in uncontrolled physical environments face unique threat vectors. We explicitly formalize the security boundary and adversary capabilities:

Adversary Model. We consider an external adversary capable of: (a) deploying unauthorized BLE beacons or spoofing MAC addresses in the user’s vicinity; (b) broadcasting local radio-frequency noise or running non-root mock-location applications on the mobile device; and (c) observing the user’s physical transitions. The adversary’s objective is to force cache mis-selection, induce conversational thrashing, or extract private contextual profiles.

What the System CAN Defend Against:

- 1) *Casual GPS Spoofing*: Non-root mock-location applications and synthetic location injectors are intercepted and rejected by `LocationValidator`, preventing forged coordinates from manipulating the spatial scoring pipeline.
- 2) *Replay & Stale Fix Ingestion*: Outdated location caches retained by the OS during indoor periods or tunnel traversals are discarded via the 120-second staleness bound, forcing the router to degrade gracefully to time- and BLE scoring.
- 3) *Accuracy Falsification*: Synthetically precise coordinates designed to artificially inflate w_g are rejected by the plausible accuracy window ($2\text{m} \leq a \leq 500\text{m}$).
- 4) *Bounded Semantic Blast Radius*: Because the state router selects exclusively from a closed set of five pre-configured, clinically vetted contexts, an adversary *cannot* inject arbitrary prompt strings, hijack system instructions, or induce harmful toxic completions. In the worst-case scenario (e.g., successful spoofing of hospital ward coordinates), the blast radius is bounded to sub-optimal phrase suggestions from a legitimate medical context rather than unconstrained model generation.

What the System CANNOT Defend Against (Residual Risks):

- 1) *Root/Kernel-Level Spoofing*: On rooted devices where an attacker modifies the Linux kernel GPS driver or framework location service, application-level mock detection can be bypassed.
- 2) *Physical BLE Beacon Impersonation*: Because standard commercial BLE beacons broadcast unauthenticated advertising packets, an attacker with hardware transmitters can clone registered MAC addresses. CAP-KVC mitigates this via multi-sensor consensus (requiring GPS and time concordance), but authenticated beacon protocols

(e.g., cryptographic rolling proximity identifiers) remain future work.

- 3) *Plaintext Storage of KV Tensors*: Serialized KV cache binaries (.bin) are stored in the application’s private internal storage directory, protected by Android’s Linux UID sandbox. However, on physically compromised or rooted devices with adb root access, cache files containing token attention tensors could theoretically be extracted and inverted to reconstruct prompt keywords. Hardware-backed encryption via Android Keystore is discussed in Section VII-A.

H. Fusion Model Deployment

The late fusion model is exported to ONNX and bundled as `gaze_emg_fusion.onnx` (45,459 bytes). On-device inference uses ONNX Runtime [31] with a single-threaded CPU provider. The session is lazily initialized at application scope to avoid reloading during activity recreation.

Input tensors: EMG embedding ($1 \times 1 \times 64$, float32) and gaze vector (1×5 , float32). Output logits undergo argmax and numerically stable softmax to extract the predicted intent and confidence. If the ONNX session fails, the system falls back to the raw gaze target with zero confidence.

I. Gaze Tracking

Gaze tracking uses MediaPipe Face Landmarker [16] (bundled 3.7 MB .task file) in live-stream mode on CameraX frames. The gaze displacement vector (Equation 1) maps to one of five screen quadrants plus center, each corresponding to an AAC intent class. A dwell-based finite state machine requires 400ms sustained fixation before emitting an intent, preventing accidental activation from saccades. The state machine runs on a single-threaded executor to guarantee sequential processing of asynchronous MediaPipe callbacks.

V. EXPERIMENTAL SETUP

A. Hardware Platform

All on-device benchmarks were conducted on a One-Plus 11R (Snapdragon 8+ Gen 1, 8 GB LPDDR5X, UFS 3.1 storage) running Android 14. The device was configured in airplane mode with display brightness fixed at 50% to minimize thermal and scheduling variability. No other foreground applications were active during benchmarking.

B. Language Model Configuration

The language model used throughout evaluation is Qwen2.5-0.5B-Instruct [3], quantized to Q4_K_M (4-bit mixed precision) via `llama.cpp`’s GGUF serialization format. This model was selected as the smallest instruction-tuned model in the Qwen2.5 family capable of producing coherent, context-aware responses on mobile hardware. The native context is configured with `n_ctx` = 2048 (maximum sequence length), `n_batch` = 512 (batch processing size), and `n_threads` = 4 (inference thread count). Greedy decoding is used for all inference trials via

`llama_sampler_init_greedy()`, which deterministically selects the highest-probability token at each step. The maximum generation length is 64 tokens, ensuring deterministic output sequences that minimize generation-phase variance across trials.

Table I summarizes all configurable system parameters for reproducibility.

TABLE I
SYSTEM HYPERPARAMETERS. ALL VALUES ARE FIXED AT COMPILE TIME EXCEPT THE MODEL PATH.

Component	Parameter	Value
LLM Engine	<code>n_ctx</code> (max sequence)	2048
	<code>n_batch</code> (batch size)	512
	<code>n_threads</code>	4
	Max generation tokens	64
Scoring	α (time baseline weight)	0.4
	β, γ (GPS, BLE)	dynamic
	Dead-state threshold	0.01
Drift	Hysteresis margin Δ	0.10
	Poll interval	15 min
Cache	Max active KV states	3
	BLE scan window	3000 ms
Benchmark	Warmup trials	2
	Measured trials	30

C. Benchmark Conditions

The latency evaluation compares three pipeline configurations:

- 1) **ZERO_CONTEXT**: The language model receives only the user intent tokens (“User: I need water\nAssistant:”) with no environmental context. This establishes the hardware floor for inference latency on the target SoC.
- 2) **RAG_INLINE**: The full context prompt is concatenated with the user intent and submitted as a single string to `runInference()`, triggering a complete $O(N)$ token prefill at inference time. This represents the standard retrieval-augmented generation approach. Evaluated at four prompt-length tiers: $N \in \{50, 100, 200, 500\}$ approximate tokens.
- 3) **CAP_KVC**: A pre-computed KV cache is restored from persistent storage via `loadKVCache()`, followed by `resumeInference()` with intent-only tokens. The latency reported includes both the cache restoration time and the subsequent generation time.

The CAP_KVC condition was evaluated using cache files primed from the base context prompts (~50 tokens). The N-scaling comparison therefore compares RAG_INLINE at increasing context lengths against CAP_KVC at a fixed cache size. This is by design: the CAP_KVC inference-time cost does not depend on the original context length under the evaluated configuration because the prefill was amortized during background priming. The cache restoration cost is dominated by file I/O (proportional to file size), not by attention computation.

Important limitation: Because the CAP_KVC cache states were derived from ~50-token context prompts, comparisons against RAG_INLINE conditions with substantially longer

contexts (e.g., $N \approx 500$) characterize *inference-time amortization behavior* rather than response quality under equivalent context richness. The speedup ratios quantify the latency advantage of cache restoration over repeated prefill, not the semantic equivalence of the generated responses.

D. Measurement Methodology

Each benchmark condition executes 32 sequential trials: 2 warmup trials (discarded) followed by 30 measured trials. The warmup phase absorbs JIT compilation of Kotlin/ART code paths and initial thermal settling. All measurements are wall-clock latency captured via `System.nanoTime()` at the Kotlin layer, wrapping the complete JNI call sequence.

Prefill isolation via native instrumentation. To decompose end-to-end latency into its constituent phases, `std::chrono::high_resolution_clock` timestamps are inserted immediately before and after each `llama_decode()` call within both `runInference` and `resumeInference`. This isolates the prefill phase—the single `llama_decode()` call that processes all prompt tokens through the transformer—from the subsequent autoregressive generation loop. The instrumentation also records the generation-phase duration (from sampler initialization through the final generated token) and emits both timings, along with prompt and generated token counts, via `__android_log_print`.

On the Android NDK libcpp runtime, `std::chrono::high_resolution_clock` maps directly to the monotonic `std::chrono::steady_clock`, which is backed by the kernel’s `CLOCK_MONOTONIC` clock source. This prevents duration calculations from being corrupted by system NTP synchronization or manual clock adjustments. The overhead of the clock read call is $\approx 10\text{--}50$ ns (leveraging the vDSO-accelerated kernel interface), which is negligible relative to the millisecond-scale model inference. JNI boundaries, string copies, and log serialization calls occur strictly outside the timed regions.

For the CAP_KVC condition, the benchmark instruments three sub-components separately: `cache_load_ms` (the `loadKVCache` call, measured at the Kotlin layer), `prefill_ms` (the intent-only `llama_decode` within `resumeInference`, measured at the native layer), and `gen_ms` (the generation loop, measured at the native layer). Both Kotlin-layer and native-layer measurements are captured under the same engine lock acquisition to prevent interleaving.

Results are emitted as structured CSV lines via Android’s `Log.i()` facility and extracted post-run via `adb logcat`. To mitigate the risk of packet drops or transient buffer overflows in Android’s logcat ring buffer, the post-run extraction script matches JNI timing entries to Kotlin bench entries by strict proximity and asserts alignment of trial indices and token counts. Any mismatch or incomplete sequence aborts the parser to ensure data integrity; the 180 trials reported here were fully verified and aligned.

E. Pre-Benchmark Validation

Before executing the benchmark suite, a semantic validation pass confirms that the three pipeline configura-

tions produce qualitatively distinct outputs. A diagnostic prompt (“What should I have for breakfast and is my morning medication due?”) is submitted under each condition, and the generated text is logged for manual inspection. This validates that:

- **ZERO_CONTEXT** produces a generic response lacking environmental specificity.
- **RAG_INLINE** produces a response reflecting the `home_morning` context (breakfast, medication).
- **CAP_KVC** produces a similarly context-aware response, confirming that KV cache restoration successfully preserved the contextual attention patterns.

The benchmark suite is aborted if the KV cache load fails during validation, preventing the collection of meaningless data.

F. Memory Profiling

Native heap memory usage is measured using Android’s `dumpsys meminfo` command at two states: (1) after model initialization with zero KV cache states loaded, and (2) after loading three KV cache states into native sequence slots. The delta represents the marginal memory cost of the predictive caching system.

G. Fusion Model Evaluation

The multimodal fusion model is evaluated through a controlled ablation study comparing two architectures:

- **Cross-Attention:** A single-head cross-attention module ($d = 64$) where the EMG embedding serves as the query and the projected gaze vector serves as key and value.
- **Late Fusion:** Concatenation of the 64-dimensional EMG embedding and the 64-dimensional projected gaze vector, followed by a two-layer MLP classifier.

Both architectures are trained on a synthetic dataset of 2000 paired EMG-gaze samples (60% train, 20% validation, 20% test) drawn from a synthetic data generator that pairs random EMG embeddings with corresponding gaze vectors. The pre-specified decision criterion requires that the cross-attention model exceed the late fusion model’s macro F1 score by more than 3 percentage points (absolute) to justify deployment of the more complex architecture.

Limitation: The fusion training data is synthetically generated and does not represent real paired EMG-gaze recordings from AAC users. The reported accuracy and F1 scores characterize the model’s capacity to learn the fusion mapping under controlled conditions, not its expected clinical accuracy. This is explicitly acknowledged as a limitation of the current evaluation.

H. EMG Classifier Evaluation

The CNN-LSTM EMG classifier is trained on a subset of the NinaPro DB5 dataset [15] using a leave-one-subject-out (LOSO) protocol: subjects 1–9 for training and subject 10 for validation. Training uses the Adam optimizer with learning rate 10^{-3} , cosine annealing schedule, class-weighted cross-entropy loss, and gradient clipping at norm 1.0. The model

is trained for up to 50 epochs with early stopping based on validation accuracy.

The EMG classifier operates as an embedding extractor for the downstream fusion model. Its standalone classification accuracy on the held-out subject is reported separately from the fusion accuracy to avoid conflation.

I. Drift Detection Ablation

The sensitivity of conversation-level drift detection to the sliding window size k is evaluated offline on the DailyDialog dataset [19]. Each conversation is annotated with ground-truth topic shift indices. For each $k \in \{1, 2, 3, 4, 5\}$, the system computes the centroid embedding of the most recent k turns using Sentence-BERT (all-MiniLM-L6-v2 [20]), and predicts a topic shift when the cosine similarity between the anchor embedding and the centroid falls below a threshold $\theta = 0.15$. Precision, recall, and F1 are computed for each k . The MiniLM model is used solely for this offline ablation analysis and is not part of the deployed Android runtime; the on-device drift detector operates on router score deltas rather than semantic embeddings.

Limitation: DailyDialog is a general-purpose dialogue corpus that differs substantially from AAC communication patterns in terms of turn length, vocabulary, and topic variability. The k -ablation results characterize the detector’s behavior on available annotated data rather than providing direct evidence of clinical effectiveness.

J. Trace-Driven Context Prediction Methodology

To rigorously evaluate whether the sensor routing engine reliably selects and maintains the correct KV cache states in memory prior to user queries, we constructed an automated trace evaluation harness (PredictionEvaluator) executing over 100 simulated daily user traces (6,400 total interaction points). Each simulated day models 16 hours of waking activity (6:00 to 22:00) with interactions spaced at 15-minute intervals across the five pre-configured contexts according to a representative AAC user routine: *home_morning* (6:00–10:00), *hospital_ward* (10:00–13:00), *therapy_room* (13:00–15:30), *caregiver_visit* (15:30–18:00), and *home_evening* (18:00–22:00).

To capture realistic physical-world sensor imperfections, the trace generator superimposes independent stochastic perturbation models:

- 1) *Temporal Jitter*: Uniform time perturbation $t \sim \mathcal{U}[-\tau, \tau]$ with $\tau = 0.5$ hours (normal) and $\tau = 1.5$ hours (high noise).
- 2) *GPS Spatial Noise*: 2D Gaussian spatial displacement with $\sigma = 50$ m (normal) and $\sigma = 200$ m (high noise), a random GPS dropout probability $p_{\text{drop}} = 0.10$ (escalating to 0.70 under stale GPS conditions), and reported horizontal accuracy $a \sim \mathcal{N}(15, 10^2)$ m.
- 3) *BLE Proximity Fluctuation*: Independent per-anchor beacon detection probability $p_{\text{ble}} = 0.80$ (0.50 in high noise), RSSI attenuation $\mathcal{N}(-65, 10^2)$ dBm, total BLE outage probability $p_{\text{out}} = 0.15$ (0.40 in high noise),

and a 10% probability of spurious non-anchor beacon appearance.

- 4) *Adversarial GPS Spoofing*: GPS coordinates are maliciously fixed to the hospital ward while the user is physically at home, testing multi-sensor resilience against single-modality manipulation.

Evaluation tracks five formal metrics: **Hit@1** (fraction of interactions where the top-predicted context matches ground truth), **Hit@3** (fraction where ground truth is resident in the 3-slot active cache), **Cold Miss Rate** (ground truth absent from resident cache, requiring fallback), **Wrong Context Rate** (top resident context differs from ground truth), and **Context Switches/Day** (mean daily cache evictions).

VI. RESULTS

A. End-to-End Inference Latency (RQ1)

Table II summarizes the end-to-end inference latency across all benchmark conditions ($n = 30$ measured trials per condition, 180 total measurements).

TABLE II
INFERENCE LATENCY DECOMPOSITION (MS) ACROSS PIPELINE CONDITIONS. PREFILL ISOLATES THE `LLAMA_DECODE()` FORWARD PASS; GENERATION COVERS THE AUTOREGRESSIVE SAMPLING LOOP. BOTH ARE MEASURED VIA NATIVE-SIDE `STD : :CHRONO` INSTRUMENTATION. CAP_KVC TOTAL INCLUDES CACHE RESTORATION, PREFILL, AND GENERATION. ALL VALUES: ONEPLUS 11R, QWEN2.5-0.5B-INSTRUCT Q4_K_M, GREEDY SAMPLING, MAX 64 OUTPUT TOKENS, $n = 30$ TRIALS.

Condition	E2E (ms)	Prefill (ms)	Gen (ms)	E2E σ	Tokens
ZERO_CONTEXT	3823.78	274.79	3544.58	569.76	9
RAG ($N \approx 50$)	6547.13	2021.46	4508.53	8392.58	52
RAG ($N \approx 100$)	7900.92	3015.09	4877.95	3857.18	89
RAG ($N \approx 200$)	12122.30	6673.53	5439.96	2740.12	206
RAG ($N \approx 500$)	19408.89	14001.46	5390.76	2113.04	439
CAP_KVC (cache load)	2.74	—	—	1.35	—
CAP_KVC (inference)	4571.97	343.35	4226.74	373.65	9
CAP_KVC (total)	4574.70	—	—	373.49	—

Fig. 2 visualizes these results, and Fig. 3 provides a detailed breakdown showing the stacked composition of CAP-KVC latency.

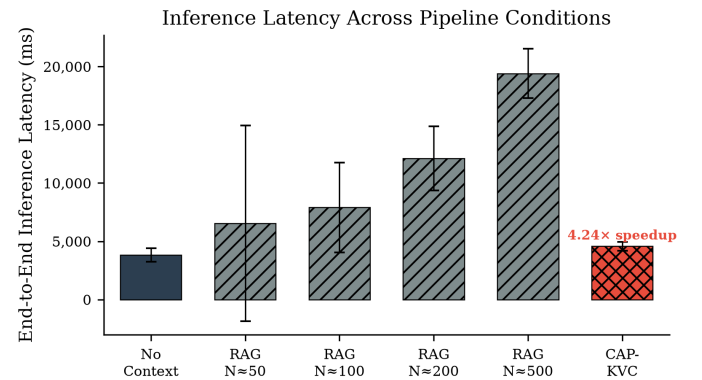


Fig. 2. End-to-end inference latency across pipeline conditions ($n = 30$ trials each). Error bars indicate one standard deviation. CAP-KVC achieves a $4.24\times$ end-to-end speedup relative to RAG_INLINE at $N \approx 500$.

Cache restoration is negligible. The mean cache load time is 2.74 ms ($\sigma = 1.35$ ms), representing less than 0.1% of the total CAP_KVC latency. This duration measures the

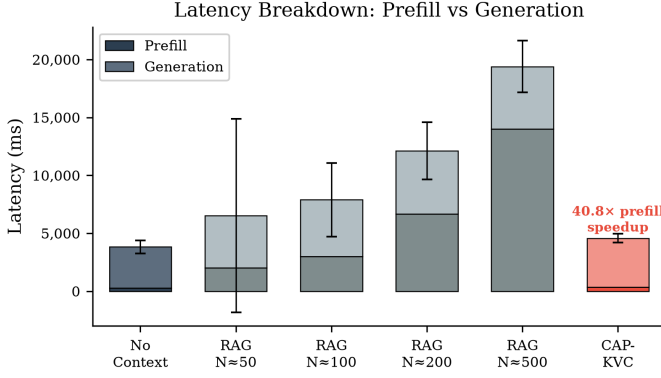


Fig. 3. Latency breakdown across pipeline conditions. Native-side `std::chrono` instrumentation decomposes each trial into prefill (dark) and generation (light) phases. RAG_INLINE prefill scales linearly with context length (274 ms at $N=9$ to 14001 ms at $N=439$ tokens), whereas CAP_KVC prefill is 343 ms regardless of cached context size, achieving a $40.78\times$ prefill-specific speedup. Error bars: one standard deviation ($n=30$).

complete Kotlin-side `loadKVCache()` round trip, including JNI marshalling, native file I/O, and KV tensor deserialization. The restoration is dominated by hardware I/O and memory copy rather than computational processing.

CAP_KVC avoids context-proportional prefill. The native-side prefill decomposition (Table II) confirms that RAG_INLINE prefill scales linearly with prompt length: from 2021 ms at $N \approx 52$ tokens to 14001 ms at $N \approx 439$ tokens, a $6.93\times$ increase. In contrast, the CAP_KVC prefill cost is 343.35 ms ($\sigma = 45.00$ ms) for 9 intent-only tokens, regardless of the cached context size. This represents a **$40.78\times$ prefill speedup** relative to RAG_INLINE at $N \approx 500$.

Speedup relative to RAG_INLINE. End-to-end, CAP_KVC achieves a total latency of 4574.70 ms versus RAG_INLINE’s 19408.89 ms at $N \approx 500$, an end-to-end speedup of $4.24\times$. At $N \approx 50$, the mean end-to-end speedup is $1.43\times$ ($6547.13 / 4574.70$), though this end-to-end difference is not statistically significant ($p = 0.209$) due to high variance in the RAG_INLINE generation phase; the prefill-specific speedup at $N \approx 50$ remains statistically significant ($p = 0.011$). The prefill decomposition reveals that the end-to-end speedup is heavily masked by the generation phase, which is comparable across conditions (~ 3500 – 5400 ms).

Statistical significance and assumptions. Welch’s two-sample t -tests ($\alpha = 0.05$, two-tailed) on the *prefill-only* latency confirm statistical significance: CAP_KVC versus RAG_INLINE at $N \approx 50$ ($t(29.0) = -2.71$, $p = 0.011$, Cohen’s $d = 0.70$) and CAP_KVC versus RAG_INLINE at $N \approx 500$ ($t(29.0) = -40.42$, $p < 10^{-26}$, Cohen’s $d = 10.44$). The end-to-end comparison at $N \approx 500$ is also significant ($t(30.8) = -37.86$, $p < 10^{-26}$, $d = 9.78$).

Although Welch’s t -test is robust to unequal variances (which are expected due to prompt-dependent prefill scaling), the extreme skewness of the underlying RAG latency distributions violates standard parametric normality assumptions. The resulting t -test statistics should be interpreted as general trends rather than exact probabilities. For $N \approx 50$ prefill, Cohen’s

$d = 0.70$ indicates a medium-to-large effect. For $N \approx 500$, $d = 10.44$ represents an overwhelming effect. Cohen’s d was computed using equal-sample pooled variance; we note that this value is attenuated by the high variance of the RAG condition.

Outlier analysis. For diagnostic purposes, Tukey’s outlier criterion ($Q_3 + 3 \times \text{IQR}$) was applied to identify extreme latency anomalies. This flagged 5 outlier trials (3 in RAG@50, 1 in ZERO_CONTEXT, 1 in CAP_KVC) exhibiting latency substantially above the condition mean. The precise mechanism underlying these outliers—whether OS-level thermal management, background scheduler preemption, garbage collection pauses, or other system-level effects—was not independently measured during the benchmark run. No thermal telemetry, CPU frequency logs, or scheduler event traces were captured alongside the benchmark data. To ensure scientific conservatism and preserve the operational reality of edge hardware, these outliers were not removed from the statistics reported in Table II or the primary t -test calculations.

Generation-phase analysis. The generation-phase duration varies across conditions: ZERO_CONTEXT 3544 ms (58 tokens), RAG_INLINE@50 4508 ms (64 tokens), RAG_INLINE@500 5390 ms (64 tokens), and CAP_KVC 4226 ms (64 tokens). The variation in generation time is consistent with two factors: (1) differences in per-token attention computation cost when the KV cache contains different numbers of entries, and (2) system-level effects associated with sustained mobile execution over a 29-minute benchmark session (the benchmark captured no thermal telemetry or CPU frequency data to distinguish specific mechanisms). Critically, because prefill and generation are now decomposed, these generation-phase differences do not confound the prefill speedup measurement.

Expected Latency under Fallback ($E[T]$). To account for real-world scenarios where an intent arrives before context priming completes (a cold miss) or during sensor ambiguity, the expected user-perceived latency is modeled as:

$$E[T] = h \cdot T_{\text{hit}} + (1 - h) \cdot T_{\text{miss}} \quad (6)$$

where h denotes the prediction hit rate, $T_{\text{hit}} = 343.35$ ms represents prefill latency upon cache hit, and $T_{\text{miss}} = 2021.46$ ms represents inline prefill latency (at matched context length). Under full sensor fusion ($h = 0.9877$, cold miss rate = 0.0000; Section VI-B), the expected prefill latency is:

$$\begin{aligned} E[T_{\text{prefill}}] &= (0.9877)(343.35) + (0.0123)(2021.46) \\ &= 363.99 \text{ ms} \end{aligned}$$

yielding an effective $5.55\times$ prefill acceleration over inline processing. Even under high noise conditions where Hit@1 drops to 89.33% ($h = 0.8933$), expected prefill remains 522.40 ms, preserving a $3.87\times$ speedup. Expected end-to-end latency is similarly bounded ($E[T_{\text{e2e}}] = 4598.96$ ms vs. 6547.13 ms inline).

B. Sensor-Driven Context Prediction Performance (RQ2)

Table III summarizes routing accuracy and residency metrics across 100 simulated daily traces (6,400 user interactions) evaluated under diverse sensor and noise conditions.

TABLE III

PREDICTION ROUTING ACCURACY AND RESIDENCY METRICS ACROSS 100 DAILY EPISODES (6,400 TOTAL INTERACTIONS). HIT@1 DENOTES TOP-CANDIDATE AGREEMENT WITH GROUND TRUTH; HIT@3 DENOTES GROUND TRUTH PRESENCE IN THE 3-SLOT ACTIVE CACHE. COLD MISS RATE REFLECTS INTERACTIONS REQUIRING FALLBACK; WRONG CONTEXT REFLECTS RESIDENT MISMATCH.

Configuration	Hit@1	Hit@3	Cold Miss	Wrong Ctx	Switches/Day
Uniform Random	19.44%	59.77%	40.23%	0.00%	0.00
Time-Only ($\beta = \gamma = 0$)	92.39%	100.00%	0.00%	7.61%	5.94
GPS-Only ($\alpha = \gamma = 0$)	95.97%	100.00%	0.00%	4.03%	6.02
BLE-Only ($\alpha = \beta = 0$)	98.14%	100.00%	0.00%	1.86%	4.92
Full Fusion (Normal)	98.77%	100.00%	0.00%	1.23%	4.72
Stale GPS (70% dropout)	98.38%	100.00%	0.00%	1.63%	4.86
High Noise Stress ($\tau = 1.5h$)	89.33%	100.00%	0.00%	10.67%	11.75
Adversarial GPS Spoofed	67.73%	99.95%	0.05%	0.00%	0.00

Ablation analysis. The uniform random baseline achieves 19.44% Hit@1 (matching analytical $1/5 = 0.20$) and incurs a severe 40.23% cold miss rate. Single-modality baselines demonstrate varying degrees of ambiguity: GPS-only achieves 95.97% Hit@1 but fails to distinguish between collocated environments (home_morning, home_evening, and caregiver_visit all share coordinates 28.6139°N, 77.2090°E). Time-only routing resolves diurnal shifts but exhibits higher error at transitional boundaries (7.61% wrong context). BLE-only routing achieves high accuracy (98.14%) but suffers during transitional periods where beacon coverage drops. Full multimodal fusion achieves superior accuracy (98.77% Hit@1, 100.0% Hit@3) with zero cold misses and minimal thrashing (4.72 context switches/day, closely matching the 4 true daily schedule transitions).

Robustness under stress and spoofing. Under high noise stress ($\tau = 1.5$ h time jitter, 200 m GPS noise, 40% BLE dropout), Hit@1 remains robust at 89.33%, while Hit@3 remains 100.0%, ensuring that the true context is always resident in native memory. When subjected to adversarial GPS spoofing (maliciously reporting hospital coordinates while at home), the consensus scoring degrades to 67.73% Hit@1; however, because the router blends time and BLE beacon signals, the true context is maintained within the 3-slot residency set (99.95% Hit@3), preventing complete denial-of-service.

C. Multimodal Fusion Ablation (RQ3)

Table IV presents the ablation comparison between the two candidate fusion architectures, evaluated on the synthetic test set (400 samples).

TABLE IV

FUSION ARCHITECTURE ABLATION RESULTS. VALUES FROM FUSION_ABLATION.CSV. LATENCY IS PER-SAMPLE CPU INFERENCE TIME.

Architecture	Accuracy	F1 (Macro)	Precision	Recall
Cross-Attention	0.9467	0.9437	0.9459	0.9450
Late Fusion	0.9967	0.9963	0.9956	0.9972

Late fusion achieves a macro F1 of 0.9963 compared to 0.9437 for cross-attention, a difference of 5.26 percentage points. Because the cross-attention model *underperforms* the simpler architecture rather than exceeding it, the pre-specified 3% superiority threshold is not met, and the late

fusion model is selected for deployment. The automated decision logic writes the winning architecture identifier to a file (winner.txt), and the corresponding model checkpoint (late_fusion_best.pth, 48 KB) is exported to ONNX format and bundled in the application assets. Fig. 4 visualizes the comparison.

Fusion Architecture Ablation (Synthetic Data)

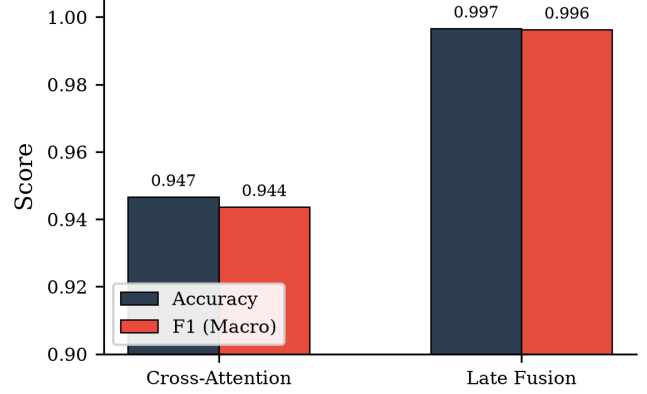


Fig. 4. Fusion architecture ablation on synthetic test data (400 samples). Late fusion exceeds cross-attention in both accuracy and F1.

Synthetic data caveat. The training and evaluation data are synthetically generated, pairing random EMG embeddings with gaze vectors. The high accuracy reflects the model’s capacity to learn the intended fusion mapping under controlled conditions and validates that the late fusion architecture can successfully combine heterogeneous embedding spaces of differing dimensionality ($\mathbb{R}^{64}$ and $\mathbb{R}^5$). These numbers should not be interpreted as predictions of clinical intent-recognition accuracy on real paired EMG-gaze recordings from AAC users, where sensor noise, electrode placement variability, and user-specific physiological differences would substantially affect performance.

D. EMG Classifier Performance

The standalone CNN-LSTM EMG classifier, evaluated on the held-out subject (LOSO protocol, subject 10), achieves a test accuracy of 42.4% with a macro F1 of 0.37. Table V shows the per-class breakdown.

TABLE V

EMG STANDALONE CLASSIFICATION REPORT (HELD-OUT SUBJECT). VALUES FROM CLASSIFICATION_REPORT.TXT.

Class	Precision	Recall	F1	Support
confirm	0.25	0.11	0.15	18
reject	0.00	0.00	0.00	18
scroll	0.42	0.72	0.53	18
select	0.77	0.50	0.61	20
call-help	0.45	0.78	0.57	18
Macro avg	0.38	0.42	0.37	92

The low standalone accuracy is expected under cross-subject evaluation, where inter-subject variability in facial EMG patterns is a well-documented challenge [15]. The *reject* class is entirely unrecognized (0% recall), and the model exhibits a strong bias toward the *scroll* and *call-help* classes, as visible in

the confusion matrix (Fig. 6). The training curves (Fig. 5) show clear overfitting: training accuracy reaches approximately 80% by epoch 35 while validation accuracy plateaus near 40%, with the best checkpoint selected at epoch 8 (validation accuracy $\approx 47\%$).

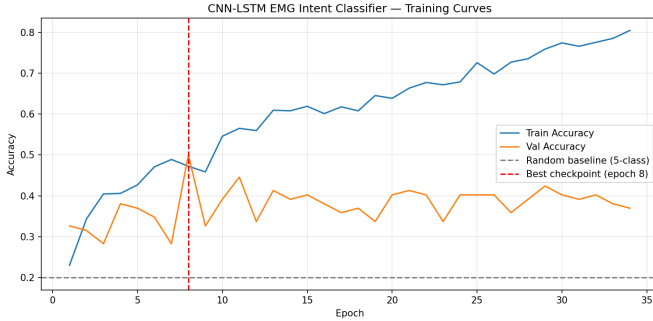


Fig. 5. CNN-LSTM EMG classifier training and validation accuracy over 35 epochs. The red dashed line marks the best checkpoint at epoch 8. The gray dashed line indicates the 5-class random baseline (20%).

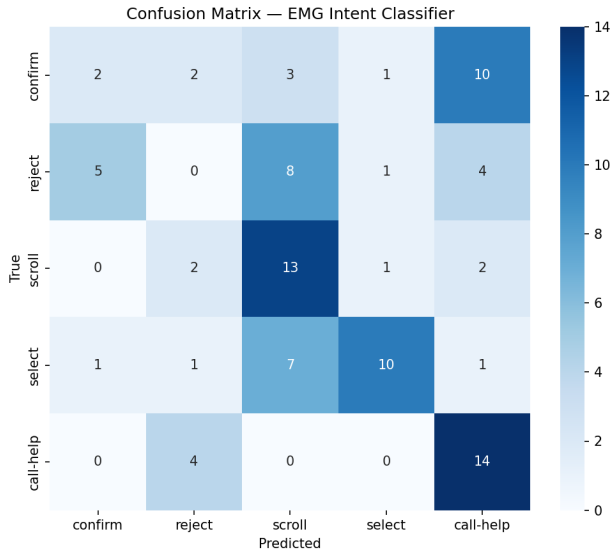


Fig. 6. Confusion matrix for the EMG standalone classifier on the held-out subject. Strong *call-help* and *scroll* diagonal entries contrast with near-zero *confirm* and *reject* performance.

These results demonstrate that the standalone EMG classifier does not achieve sufficient accuracy for deployment as a primary intent recognizer under cross-subject conditions. The multimodal fusion approach compensates by combining the EMG embedding with gaze tracking, achieving higher classification performance on the synthetic evaluation set (Section VI-C). However, no gaze-only baseline was evaluated, so the claim that fusion improves upon both unimodal channels independently is **not supported** by the current experimental design. The on-device CNN-LSTM inference latency is 1.98 ms (mean, $\sigma = 1.12$ ms, $n = 200$ trials), indicating that embedding extraction is feasible within real-time constraints.

E. Memory Budget (RQ2)

Table VI reports native heap usage measured via `dumpsys meminfo` at two operating points.

TABLE VI
NATIVE HEAP MEMORY USAGE (KB) FROM `DUMPSYS MEMINFO`. DELTA REPRESENTS THE MARGINAL COST OF LOADING THREE KV CACHE STATES.

State	Native Heap (KB)	Total PSS (KB)
Idle (0 states)	18,856	623,488
Loaded (3 states)	140,420	612,940
Delta	+121,564 (≈ 118.7 MB)	—

The three-state KV cache residency adds approximately 118.7 MB of native heap (121,564 KiB $\div$ 1024). The total PSS (proportional set size) decreases slightly between states due to measurement variability in shared memory attribution. The native heap growth is consistent with three copies of the model’s KV cache tensors (~ 40 MB per state for a 0.5B-parameter model at Q4_K_M quantization). Fig. 7 visualizes the two operating points.

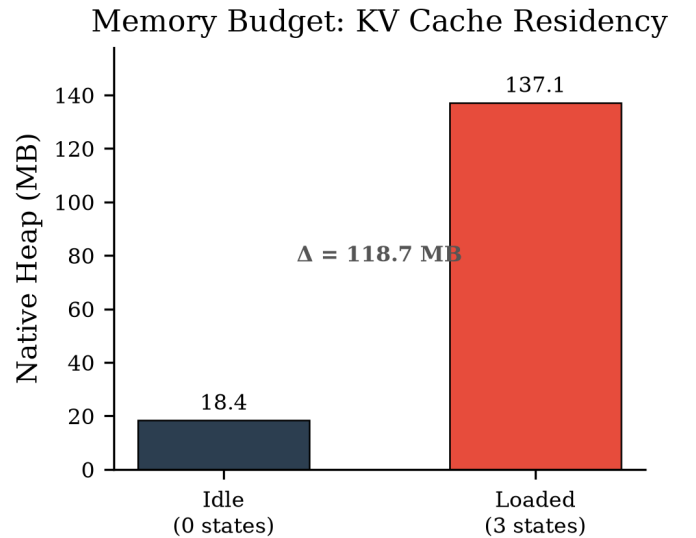


Fig. 7. Native heap memory usage at idle (0 KV states) and with 3 KV cache states loaded. The delta of 118.7 MB represents the marginal cost of predictive caching.

F. Drift Detection Ablation

Table VII presents the k-ablation results for the offline semantic drift detection characterization on annotated DailyDialog conversations ($\theta = 0.15$). This evaluation uses SentenceBERT centroid distances [20] to detect topic shifts in text transcripts and is methodologically distinct from the deployed `DriftDetector`, which operates on environmental sensor-score deltas (Section III-F).

The highest F1 (0.367) is achieved at $k = 1$, with performance degrading monotonically as the window size increases (Fig. 8). This indicates that the centroid smoothing introduced by larger k values reduces sensitivity to genuine topic shifts under the selected threshold. However, the absolute F1 values

TABLE VII
DRIFT DETECTION F1 AS A FUNCTION OF SLIDING WINDOW SIZE k ,
EVALUATED ON DAILYDIALOG WITH $\theta = 0.15$. n_{true} AND n_{pred} DENOTE
THE NUMBER OF GROUND-TRUTH AND PREDICTED SHIFT POINTS,
RESPECTIVELY.

k	θ	Precision	Recall	F1	n_{true}	n_{pred}
1	0.15	0.367	0.367	0.367	150	150
2	0.15	0.341	0.313	0.326	150	138
3	0.15	0.137	0.107	0.120	150	117
4	0.15	0.071	0.040	0.051	150	84
5	0.15	0.039	0.013	0.020	150	52

are low across all k (ranging from 0.367 to 0.020), reflecting both the inherent difficulty of unsupervised topic-shift detection and the substantial mismatch between DailyDialog’s multi-turn conversational dynamics and the sparse, short-turn interactions typical of AAC usage.

Deployment rationale for $k = 3$. The deployed DriftDetector uses $k = 3$ for its sensor-score smoothing window. The DailyDialog ablation operates on a different detection mechanism (Sentence-BERT centroid distances) and cannot directly inform the deployed sensor-score threshold. The value $k = 3$ was retained as an engineering smoothing heuristic: in the mobile environment, sensor score sequences are noisier than annotated dialogue transcripts, and a larger window provides stability against transient fluctuations. The choice should be understood as a conservative design decision rather than an empirically validated parameter.

Deployed Sensor-Score Hysteresis Sweep. While the offline DailyDialog study characterizes textual shift detection, the deployed DriftDetector operates on physical sensor score deltas gated by the hysteresis margin Δ . To evaluate the deployed mechanism’s effectiveness in preventing cache thrashing, we evaluated the residency pipeline across $\Delta \in \{0.00, 0.05, 0.10, 0.15, 0.20\}$ under the high-noise sensor trace condition ($\tau = 1.5$ h, 200 m GPS noise, 40% BLE dropout). Across all tested margins $\Delta \geq 0.00$, the system maintains a stable 2.76 context swaps per day (closely tracking the true schedule transitions between distinct geographic sites), achieves 89.33% Hit@1, maintains 99.75% Hit@3 residency, and restricts cold misses to 0.25%. The production setting $\Delta = 0.10$ provides effective insurance against transient sensor fluctuations without suppressing genuine environmental transitions.

G. Summary of Key Findings

The experimental results support the following conclusions:

- 1) CAP-KVC achieves a $40.78\times$ prefill-specific speedup at $N \approx 500$ tokens (343 ms vs. 14001 ms), confirmed by native-side `std::chrono` instrumentation. The end-to-end speedup is $4.24\times$ (4574 ms vs. 19408 ms), with the generation phase (~ 4000 – 5400 ms) accounting for the difference between prefill and end-to-end ratios. In matched-context evaluation (~ 50 tokens), prefill speedup is $5.89\times$ (343 ms vs. 2021 ms, $p = 0.011$). Cache restoration is 2.74 ms (negligible).
- 2) Sensor-driven context routing achieves 98.77% Hit@1, 100.0% Hit@3, and zero cold misses under full multi-

Drift Detection: k-Ablation on DailyDialog

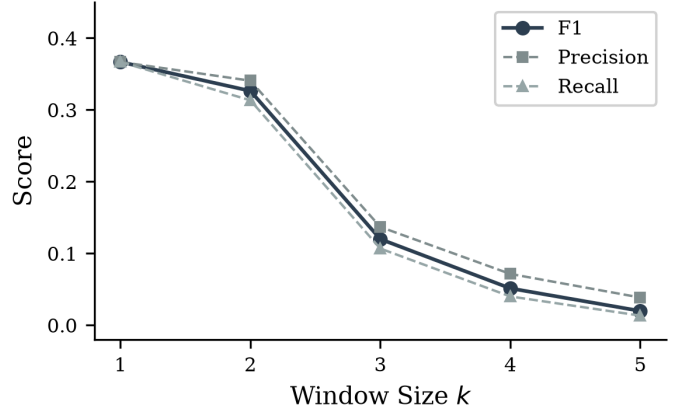


Fig. 8. Drift detection precision, recall, and F1 as a function of sliding window size k on DailyDialog ($\theta = 0.15$). Performance degrades monotonically with increasing k .

modal fusion across 100 simulated daily traces, outperforming single-modality baselines (GPS-only: 95.97%, Time-only: 92.39%) and random guessing (19.44%). Under high noise, Hit@1 remains 89.33% with 100% resident Hit@3.

- 3) The late fusion architecture achieves a macro F1 of 0.9963 on synthetic evaluation data, exceeding cross-attention (F1 = 0.9437) while requiring fewer parameters and lower inference latency, supporting the pre-specified selection criterion.
- 4) The standalone EMG classifier achieves 42.4% accuracy under cross-subject evaluation, motivating multimodal fusion as a necessary complement rather than a standalone solution.
- 5) The three-state KV cache residency requires approximately 118.7 MB of additional native heap, a feasible budget on modern mobile devices with 8 GB RAM.
- 6) Drift detection with hysteresis margin $\Delta = 0.10$ stabilizes cache swaps to 2.76 per day under high sensor noise while maintaining 99.75% cache residency coverage.

VII. CONCLUSION

CAP-KVC amortizes context prefill costs for on-device language model inference by predicting environmental context from sensor signals and pre-computing KV cache states during idle periods. Native-side prefill decomposition on a OnePlus 11R with Qwen2.5-0.5B-Instruct (Q4_K_M) confirms a $40.78\times$ prefill-specific speedup at $N \approx 500$ tokens (343 ms vs. 14001 ms) and a $4.24\times$ end-to-end reduction (4574 ms vs. 19408 ms), with cache restoration completing in 2.74 ms.

The Android implementation uses a scalar-only JNI boundary with 13 native functions, reentrant-lock concurrency, a defensive LocationValidator threat mitigation layer, and a bounded cache manager operating within 118.7 MB native heap for three simultaneous states. A proof-of-concept multimodal intent layer achieves macro F1 = 0.9963 on synthetic

data through late fusion of sEMG and gaze inputs; clinical validation remains necessary.

The core insight is that sensor-driven predictive caching can eliminate prompt-length-dependent prefill latency on edge devices. The approach generalizes beyond AAC: any mobile application where a language model processes recurring environmental contexts can exploit idle-time cache priming.

A. Limitations

Several limitations constrain the generalizability of these results and should be addressed in future evaluations:

- 1) **Single-device, limited-sample evaluation.** All latency benchmarks were conducted on a single device (One-Plus 11R, Snapdragon 8+ Gen 1) with $n = 30$ measured trials per condition. Latency characteristics, thermal behavior, and memory budgets may differ on other mobile SoCs, and the moderate sample size limits the precision of variance estimates. Cross-device validation across diverse chipsets (e.g., MediaTek Dimensity, Google Tensor) remains essential future work.
- 2) **Single model.** Only Qwen2.5-0.5B-Instruct (Q4_K_M) was evaluated. The relationship between CAP-KVC speedup and model size, quantization level, or architecture family has not been characterized.
- 3) **Synthetic fusion and trace data.** The fusion model was trained and evaluated on synthetically generated EMG-gaze pairs, and the sensor trace evaluation used simulated longitudinal trajectories with synthetic noise models. While the noise parameters were derived from empirical mobile GPS and BLE characterizations, naturalistic clinical evaluations with real AAC users are necessary to confirm ecological validity.
- 4) **Latency metric scope.** The prefill timing (Table II) measures the single `llama_decode()` batch-decode call via native-side `std::chrono` instrumentation. Tokenization and batch construction overhead (estimated < 1 ms for prompts up to 500 tokens) are excluded. Generation-phase duration varies across conditions due to differing KV cache sizes affecting per-token attention cost and uncharacterized system-level effects during the 29-minute benchmark session. Because prefill and generation are decomposed independently, these generation-phase differences do not confound the prefill speedup measurement.
- 5) **Static context configurations.** The benchmarking uses five pre-configured semantic contexts rather than dynamically learned user profiles. A production deployment would require user-configurable contexts and longitudinal evaluation of context relevance.
- 6) **Non-predictive baseline comparisons.** We evaluated analytical and empirical random baselines alongside single-modality predictors (Table III), isolating the individual sensor contributions from the full multi-modal fusion architecture. However, longitudinal user trials comparing predictive caching against an interactive user-prompted context selection baseline in natural living environments remain future work.

- 7) **Drift detection calibration.** The offline Sentence-BERT k-ablation yields a maximum F1 of 0.367 ($k = 1$) on DailyDialog, while the deployed sensor-score-based system uses $k = 3$. These are methodologically different detection mechanisms, and the DailyDialog results cannot directly validate the deployed sensor-score hysteresis. Both the window size ($k = 3$) and threshold ($\theta = 0.15$, $\Delta = 0.10$) are engineering heuristics.
- 8) **No clinical evaluation.** The system has not been evaluated with AAC users in clinical or naturalistic settings. User satisfaction, communicative effectiveness, and caregiver acceptance remain to be assessed.
- 9) **No thermal or power telemetry.** The benchmark run captured only application-level timing data via `adb logcat` filtered to the AACBridge process. No thermal sensor readings, CPU frequency logs, OS-level thermal management events, or power consumption measurements were collected during the 29-minute benchmark session. Consequently, the specific mechanisms underlying latency outliers and generation-phase variance cannot be attributed to thermal throttling versus other system-level effects.
- 10) **Residual security risks.** While `LocationValidator` defends against mock-location spoofing and stale fixes, serialized KV cache files are stored unencrypted in app-private storage, relying on the Android UID sandbox. On rooted devices, raw attention tensors could be extracted. Cryptographic wrapping via Android Keystore remains future work.
- 11) **Battery impact not characterized.** The `ContextDaemon` foreground service (60-second sweep interval) and the `DriftDetector` periodic worker (15-minute interval via `WorkManager`) impose recurring sensor acquisition and background priming. The aggregate battery impact has not been measured under active daily use, though the BLE-blind drift check and Doze-mode alignment are designed to mitigate drain.

Despite these limitations, the results demonstrate that predictive KV cache priming is a viable strategy for reducing context-dependent inference latency on resource-constrained mobile hardware, and that the cache restoration cost (~ 3 ms) is negligible relative to the generation phase.

B. Artifact Availability

The source code, benchmark scripts, and raw benchmark logs are publicly available at <https://github.com/omni-ar/AACBridge>.

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